

Computer Science - Advising Guide 2025/2026

Total Credits - 123

**** DISCLAIMER** - This is a resource put together from the information on the 2025-2026 Virginia Tech Catalog, the 2024/2025 Roadmap, and the 2023/2024 Checksheet (semester offerings only) for Computer Science. Please utilize the Virginia Tech Catalog and the registration systems in Hokie Spa to confirm this information. ******

First Year					
FALL			SPRING		
Course	Course Title	Credits	Course	Course Title	Credits
CS 1114	Intro to Software Design (C) <i>Co: MATH 1225</i>	3	CS 2114	Software Design & Data Struct. (C) <i>Pre: CS 1114 or CS 2604</i>	3 [F,S,SI,II]
ENGE 1215	Foundations of Engineering	2	ENGE 1216	Foundations of Engineering <i>Pre: ENGE 1215</i>	2
ENGL 1105	First-Year Writing	3	ENGL 1106	First-Year Writing <i>Pre: ENGL 1105</i>	3
MATH 1225	Calculus of a Single Variable <i>Pre: MATH Placement (ALEKS)</i>	4	MATH 1226	Calculus of a Single Variable <i>Pre: MATH 1225 (C-)</i>	4
	Natural Science Elective	4		Natural Science Elective	4
Total Credit Hours		16	Total Credit Hours		16

Second Year					
FALL			SPRING		
Course	Course Title	Credits	Course	Course Title	Credits
CS 1944	CS First Year Seminar <i>Pre: CS 1114 or CS 2064 or ECE 2514</i>	1 [F,S]	CS 2506	Intro to Computer Org. (C) <i>Pre: (CS 2114 or ECE 2514), (CS 2505 or ECE 2564), (MATH 2534 or MATH 3034)</i>	3 [F,S]
CS 2104	Intro to Problem Solving in CS (C) <i>Pre: MATH 1225, (CS 1114 or CS 2064 or ECE 2514)</i>	3 [F,S,SI]	MATH 2114	Intro to Linear Algebra <i>Pre: MATH 1225 (B) or MATH 1226</i>	3
CS 2505	Intro to Computer Org. (C) <i>Pre: CS 2114; Co: MATH 2534 or MATH 3034</i>	3 [F,S,SI]	MATH 2204	Intro to Multivariable Calculus <i>Pre: MATH 1226</i>	3
MATH 2534	Intro to Discrete Mathematics <i>Pre: CS 1114 or ECE 1004 or CS 2064 (MATH Double Majors take MATH 3034)</i>	3 [F,S,S1]		Communications Elective	3
	Pathways 2, 3, or 6a	3		Advanced Natural Science Elective	4
	Pathways 2, 3, or 6a	3			
Total Credit Hours		16	Total Credit Hours		16

Third Year					
FALL			SPRING		
Course	Course Title	Credits	Course	Course Title	Credits
CS 3114	Data Structures & Algorithms (C) <i>Pre: (CS 2114 or ECE 3514), (CS 2505 or ECE 2564), (MATH 2534 or MATH 3034)</i>	3 [F,S,SI]	CS 3214	Computer Systems <i>Pre: (CS 2506, CS 2114) or (ECE 2564, ECE 3574)</i>	3 [F,S]
	CS 3/4/5XXX Elective	3	CS 3604	Professionalism in Computing <i>Pre: CS 1944, (CS 2114 or ECE 3514), (COMM 2004 or COMM 2014)</i>	3 [F,S]
MATH 3134	Applied Combinatorics <i>Pre: MATH 1226, (MATH 2534 or MATH 3034)</i>	3		Statistics Elective	3
	Professional Writing Elective	3		Pathways 2, 3, or 6a	3
	Pathways 2, 3, or 6a	3		Pathways 2, 3, or 6a	3
ENGE 3900	Bridge Experience	0			
Total Credit Hours		15	Total Credit Hours		15

Fourth Year					
FALL			SPRING		
Course	Course Title	Credits	Course	Course Title	Credits
CS 3304	Comparative Languages <i>Pre: CS 3114</i>	3 [F,S]	CS 4944	Seminar <i>Pre: CS 3604</i>	1 [F,S]
	CS Theory Elective	3	CS 4094	Computer Science Capstone <i>Pre: CS 2505, CS 3114</i>	3
	CS 3/4/5XXX Elective	3		CS 4/5XXX Elective	3
	CS Technical Elective	3		Free Elective	3
	Free Elective	3		Free Elective	4
Total Credit Hours		16	Total Credit Hours		14

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CS Department Elective Requirements

- Pathways to General Education

Pathways 1f (foundational) Discourse	ENGL 1105	3	ENGL 1106	3
Pathways 1a (advanced) Discourse	Communications Elective, Professional Writing Elective, or Free Elective			3
Pathways 2 Critical Thinking in the Humanities	<i>Student chooses the course</i>	3	<i>Student chooses the course</i>	3
Pathways 3 Reasoning in the Social Sciences	<i>Student chooses the course</i>	3	<i>Student chooses the course</i>	3
Pathways 4 Reasoning in the Natural Sciences	Natural Science Elective	4	Natural Science Elective	4
Pathways 5f (foundational) Quantitative & Computational Thinking	MATH 1225	4	MATH 1226	4
Pathways 5a (advanced) Quantitative & Computational Thinking	CS 3114			3
Pathways 6a (arts) Critique & Practice in the Arts	<i>Student chooses the course</i>			3
Pathways 6d (design) Critique & Practice in Design	ENGE 1215 + ENGE 1216 (or ENGE 1414)			4

- See the [Virginia Tech Catalog](#) for information about Pathways options.
- As of October 1, 2025, **Pathways concept 7**, Critical Analysis of Identity and Equity in the U.S., has been suspended. All students at Virginia Tech no longer need to complete this requirement.
 - A Free Elective (3-credits) was added to this plan to account for losing the credit hours originally included for the Pathways 7 course.
 - Students who have already completed a Pathway 7 course can count those credit hours towards the newly added Free Elective requirement.
- Natural Science Elective (Choose 2 of the following - total of 8 credits)
 - Biology
 - BIOL 1105 - Principles of Biology (3 credits) [F,SI]
 - BIOL 1115 - Principles of Biology Lab (1 credit) [F,SI]
 - Co: BIOL 1105
 - Chemistry
 - CHEM 1035 - General Chemistry (3 credits)
 - Pre: CHEM Placement (ALEKS)
 - CHEM 1045 - General Chemistry Lab (1 credit)
 - Co: CHEM 1035
 - Physics
 - PHYS 2305 - Foundations of Physics (4 credit lecture, 0 credit lab)
 - Pre: MATH 1225; Co: MATH 1226
- Advanced Natural Science Elective (Choose 1 of the following - total of 4 credits)
 - Biology
 - BIOL 1106 - Principles of Biology (3 credits) [S,SII]
 - BIOL 1116 - Principles of Biology Lab (1 credit) [S,SII]
 - Co: BIOL 1106
 - Chemistry
 - CHEM 1036 - General Chemistry (3 credits)
 - Pre: CHEM 1035
 - CHEM 1046 - General Chemistry Lab (1 credit)
 - Pre: CHEM 1045; Co: CHEM 1036

Computer Science - Advising Guide 2025/2026

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- Physics
 - PHYS 2306 - Foundations of Physics (4 credit lecture, 0 credit lab)
 - Pre: PHYS 2305, MATH 1226
- Communications Elective (Choose 1 of the following)
 - COMM 2004 - Public Speaking (3 credits)
 - Pre: Sophomore Standing
 - COMM 2014 - Speech Communication (3 credits)
 - Pre: Sophomore Standing
 - **Note:** COMM 2004 Public Speaking can be used to satisfy Pathways 1A. Students who do not take COMM 2004 Public Speaking as their communications elective will need to satisfy Pathways 1A through a suitable professional writing elective or free elective.
- Professional Writing Elective (Choose 1 of the following)
 - ENGL 3764 - Technical Writing (3 credits)
 - Pre: ENGL 1106, Junior Standing
 - ENGL 3804 - Technical Editing and Style (3 credits)
 - Pre: ENGL 1106
 - ENGL 3814 - Creating User Documentation (3 credits)
 - Pre: ENGL 1106
 - ENGL 3824 - Visual Rhetoric and Document Design (3 credits)
 - Pre: ENGL 1106
 - ENGL 3834 - Intercultural Issues in Professional Writing (3 credits)
 - Pre: ENGL 1106
 - ENGL 3844 - Writing and Digital Media (3 credits)
 - Pre: ENGL 1106
 - ENGL 4824 - Science Writing (3 credits)
 - Pre: ENGL 1106
 - **Note:** ENGL 3764 Technical Writing can be used to satisfy Pathways 1A. Students who do not take ENGL 3764 Technical Writing as their communications elective will need to satisfy Pathways 1A through a suitable communications elective or free elective.
- Statistics Elective
 - See the [Virginia Tech Catalog](#) for information about the elective options.
- CS Theory Elective
 - See the [Virginia Tech Catalog](#) for information about elective options.
- CS Technical Elective
 - See the [Virginia Tech Catalog](#) for information about elective options.
- CS 3/4/5XXX Elective
 - See the [Virginia Tech Catalog](#) for information about elective options.
- CS 4/5XXX Elective
 - See the [Virginia Tech Catalog](#) for information about elective options.
- Free Electives
 - Any course that does not meet a specific graduation requirement.
- CS Non-Technical Course Requirement
 - B.S. in CS students must complete 30 credits of non-technical courses. All courses are approved as non-technical courses except those in the departments of Biological Sciences, Chemistry, Geosciences, Physics, Mathematics, and Statistics, and all departments in the College of Engineering, except for engineering courses satisfying Pathways 7. Also excluded are courses listed as CS technical electives.
- Common Substitutions
 - ENGE 1414 (4 cr) may be substituted for ENGE 1215 (2 cr) + ENGE 1216 (2 cr)
 - MATH 2405H (5 cr) + MATH 2406H (5 cr) may be substituted for MATH 2114 (3 cr) + MATH 2204 (3 cr) + free elective (4 cr)
 - CS 2064 (C) (3 cr) may be substituted for CS 1114 (C) (3 cr)
 - ECE 2514 (C) (3 cr) may be substituted for CS 1114 (C) (3 cr)
 - ECE 3514 (C) (3 cr) may be substituted for CS 2114 (C) (3 cr)
 - ECE 2564 (C) (3 cr) may be substituted for CS 2505 (C) (3 cr)